



The LHC

What is the LHC?

The LHC, or Large Hadron Collider, is the world's largest particle accelerator, which will make proton beams collide with each other at nearly the speed of light.

How does this matter?

Scientists believe that by colliding these beams, they will be able to verify the existence of the Higgs Boson, a particle that might tell us more about how the universe started. Other scientists—notably Stephen Hawking—believe that all we will know is that there is no Higgs Boson.

Why is it popular?

The LHC has been in the news for many reasons, the most significant being the possibility that when activated, it will end the universe. It will create "big-bang-like" conditions, and to some, that means that we'll all vanish in a big bang. This was supposed to happen in October, but due to a malfunction (that will cost around \$21 million to repair, incidentally), the LHC is now offline till next summer.

Where does one put a particle accelerator?

This particular accelerator occupies an underground tunnel, which has a circumference of 27 km, at the border between France and Switzerland.

Who is behind this madness?

The LHC is a joint effort by scientists from across the world - 10,000 scientists from over 100 countries, to be more specific - under the umbrella of the European Organisation for Nuclear Research (CERN).

When did all this come about?

The LHC was approved way back in 1995, and was supposed to have been completed by 2005. That became April 2007, which finally became September 2008.

executives could get a much-needed \$440,000 (Rs 2.06 crore) vacation. Some others, however, decided that all the bailout begging is beneath them, so they took matters into their own dirty hands.

Turns out that the top three LCD manufacturers in the world — LG, Sharp and Chunghwa Picture Tubes — belong to the latter group. Early in November, they pled guilty to a price-fixing scam — they've all been selling LCD panels at much higher prices than they're really worth. Apparently, the companies hated that the price of LCD panels were dropping at such an alarming rate, so they decided to slow that decline to keep their earnings up.

If you've been watching LCD prices lately, you'll have noticed that they've continued to fall at an alarming rate, even with the price-fixing madness. According to iSuppli, a market research firm, the price of a 15.4-inch laptop screen has dropped from \$97 (Rs 4,600) to \$63 (Rs 2,961) in just the last six months. And remember, this is after the LCD makers have played their mischief. Take away the price fixing, and we're talking much lower prices for your TVs, monitors and laptops. Or, at least, a bigger profit for the companies that sell you these products.

For their indiscretions, the three companies will receive more than just a rap on the knuckles. The US Justice Department has fined them a total of \$585 million (Rs 2750 crore); LG alone will have to cough up \$400 million (Rs 1880 crore). Obviously, an irate



American public won't stand for being gypped (not when there's money to be made), and class action lawsuits have cropped up all over the landscape. It's also likely that the companies who have been buying these LCDs — Dell, Apple and Motorola, for instance — might want some revenge.

MOVING PICTURES

Photos That Follow The Sun

Even though we've gone from the intimate detail of the film photo to the grainy disappointment of the digital photo to the intimate detail of the multi-megapixel (gigapixel, even) photo, the photograph itself hasn't changed. It captures just a single moment in time, and remains the same forever.

Now, Martin Fuchs at the Max Planck Institute for Informatics wants to change that. Instead of having you look at an unchanging photo, he wants you to see it evolve over the day. Despite that, as you look at it, you'll see shadows move, lights

come on, and other such goodness that you're only used to in movies.

The setup comprises a lens array and focuses light on a transparent photo film. You see the image on a diffuser. The photograph on the film is really a composite of hundreds of photographs taken all through the day.

When the light from the lens array falls on a hexagon, you'll see only one source image at a given time. As the sun moves behind the photo, a different photo is illuminated, so it looks like the photo is changing with time.

It's a great idea for billboards, but we doubt we'll be taking any such photos for our personal collections. Firstly, there's the matter of being patient enough to wait while your camera spends all day taking pictures. Secondly, the procedure doesn't seem like one that your local photo shop can accomplish. And finally, the fact that the sun needs to be behind the photo restricts our choices of placement. Designer window? Yes, that seems to be an option with this innovation. Photo frame on the coffee table, probably not. Still, 'twould be a great idea to break away from just any other photo. ■